

ZS-M44

The Conditional Register-Trace Normalization of the Z-Spin Block-Laplacian and Its Regge-Cell Realization: Exact Removal of a Spurious Sector Asymmetry and the Open Regge-Hessian Gate

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Theme / Code: Math Spine / ZS-M44 · Companion to ZS-M6 · Upstream of ZS-F36

Verification: 19/19 PASS | **Zero fitted parameters** | corrects the a_0/a_1 conflation; establishes the

normalized-trace bridge $\kappa^2 = A, \langle a | \rho_Q | a \rangle = A/Q$ (with $\rho_Q = I_Q/Q$) **DERIVED-CONDITIONAL** on that register-scalar density; integrates ZS-M45's Regge-cell lemma (§6), whose global operator $\hat{R} \propto I_Q$ is **OPEN**; the R-2 **perturbation-artifact correction** (§10.4 0.855 $\approx \delta_x/\delta_y$ trap) stands as the firm result | **(A, Q, dim Z) = (35/437, 11, 2) LOCKED**.

§0. Abstract

ZS-M6 proved $\kappa^2 = A/Q = 35/4807$ under three honest caveats — **R-1** (the 1-loop Regge-lattice derivation was heuristic), **R-2** (the register-scalar assumption), **R-3** (rank-1 from the action). This paper's firm result is narrower and honest: it **exactly removes a spurious sector-asymmetry artifact** (PROVEN), reduces $\kappa^2 = A/Q$ to a single **register-trace normalization** ($\rho_Q = I_Q/Q$;

DERIVED-CONDITIONAL), and — via the integrated ZS-M45 Regge-cell lemma and a **graph-Laplacian coordinate diagnostic** (§6.3) — shows the local cell deficit is sector-uniform while the **global register-scalar operator** $\hat{R} \propto I_Q$ **remains OPEN**. The diagnostic (coordinate Rayleigh quotients 0.600 vs 0.244; graph eigenvalue triples $2 - \sqrt{2}$ vs 0.243) gives *evidence* the sector diagonals differ, but is **not** the genuine $B^\top W_{\text{hinge}} B$ Regge Hessian; it narrows the residual to the action-level selection of a register measure (mode-count and metric being two benchmarks), with the a_0 heat-trace coefficient arguing for the democratic mode-count reading but not deriving it.

R-1 (Register-Trace Normalization, M44.T1) [DERIVED-CONDITIONAL]. In the finite-matrix heat trace $\text{Tr}, e^{-tL} = a_0 - a_1 t + \dots$, the *mode count* is $a_0 = \dim = Q$ (leading rank coefficient) — **not** $a_1 = \text{Tr}, L$ (which equals 13.30 $\neq Q$). The per-mode normalization is the **normalized trace**

$\tau_Q = Q^{-1} \text{Tr}$. With the Gilkey product coupling $A = \delta_x \delta_y$ (PROVEN) and the assumption that A distributes via the maximally-mixed register density $\rho = I_Q/Q$, the per-mode cross-coupling is

$\kappa^2 = A, \tau_Q(\rho - \text{projected}) = A/Q$. The value is correct; but $\rho = I_Q/Q$ is *asserted* (as the symmetry-natural / democratic state), **not derived from the action** — so R-1 is

DERIVED-CONDITIONAL on this register-trace normalization, and merges with R-2. (A separate,

genuine fact from ZS-M6 §4.3: $a_1 = \text{Tr}, L$ is coupling-immune, since V is off-diagonal — this preserves the *mode count* but does not by itself fix the democratic *distribution*.)

R-2 (democratic normalization). The register-scalar assumption splits into (a) **within-irrep** constancy and (b) **cross-sector** democratic normalization. R-2(a) is DERIVED: a G -equivariant coupling is a scalar on each irrep (Schur), and all three targets are dimension-3 irreps of the polyhedral symmetries (O_h for X , I_h for Y). R-2(b) is **constrained** by an **exact arrowhead inverse-eigenvalue solve** of the ZS-M6 §2.3 spectrum (a *consistency* check, not an action-independent forward derivation): the recovered couplings are $g_X^2/\kappa^2 = 2.996$, $g_{Y_1}^2/\kappa^2 = 2.989$, $g_{Y_2}^2/\kappa^2 = 3.021$ — **democratic to 0.45%**

(input-precision-limited), decisively excluding the sector-weighted alternative ($g_X^2/g_Y^2 = \delta_X/\delta_Y = 0.865$, a 13.5% gap). The exact solve also **corrects ZS-M6 §10.4**: the reported ratios (2.61, 3.05, 3.13) are an artifact of the 2nd-order formula $g^2 = \Delta\lambda(\lambda - \lambda_{\beta_0})$, which happens to mimic δ_X/δ_Y — a genuine trap that could be misread as sector-weighting.

R-3 is DERIVED-CONDITIONAL (on mediator Hom-multiplicity-one) from ZS-M6 §7A (all-orders $L_{XY} = 0 + \text{Schur}$); the intertwiner multiplicities $\dim\text{Hom}_G(V_X, V_Z)$, $\dim\text{Hom}_G(V_{Z'}, V_Y)$ are not computed here.

Result (honest). The firm result is the **R-2 perturbation-artifact correction** (§5.4): the ZS-M6 §10.4 “~15% asymmetry” is a 2nd-order-perturbation artifact spuriously mimicking δ_X/δ_Y ; the exact arrowhead solve gives democratic $g_r^2/\kappa^2 \approx 3$. The **register-scalar normalization** ($\kappa^2 = A/Q$, and $g_{reg}^2 = 6A/Q$) is **DERIVED-CONDITIONAL on register-trace normalization** — the assumption that A distributes via $\rho = I/Q$, which R-2(a) Schur constrains within each irrep but which is not derived from the action across sectors. The arrowhead inversion (§5.3) and the 50-digit recompute (§5.5) are **consistency checks** (they use, or reconstruct with, the democratic ansatz), *not* a fully action-independent forward derivation. Consequently the ZS-F36 $\tilde{G}_s = 1$ normalization is **DERIVED-CONDITIONAL on register-trace normalization**, not “discharged.” The metric-scale no-go (M_{UV}) is untouched. Verification 19/19 + 14/14 PASS.

Epistemic Status Legend

Tag	Meaning
PROVEN	Explicit proof or exact machine verification; no undischarged assumption.
DERIVED	Follows from PROVEN results by stated steps; no new parameter.
DERIVED-BY-INHERITANCE	Uses an upstream corpus result, not re-proven; inherits its (now-discharged) caveats.
CONFIRMED-NUMERICALLY	Established by an exact computation to a stated, input-limited precision.
IMPORTED-PROVEN / STANDARD	Established externally (Gilkey heat-kernel theory, Schur’s lemma).
IDENTITY	Arithmetic consistency; no new physical content.
OPEN	A genuine remaining gap.

§1. Introduction

ZS-M6 constructed the 11×11 register Block-Laplacian $L = L_X \oplus L_Z \oplus L_Y + C_{XZ} + C_{ZY}$ with rank-1, β_0 -selected cross-coupling $\kappa = \sqrt{A/Q}$, proved $\kappa^2 = A/Q$ (Register-Total Normalization) and $g_\Gamma^2 = \dim(\Gamma)\kappa^2$ (Dimensional Coupling Norm), and flagged **R-1** (heuristic 1-loop normalization), **R-2** (register-scalar assumption), **R-3** (rank-1 from the action). ZS-M44 v1.0–v1.1 addressed these; **the current version corrects the a_0/a_1 conflation and honestly re-scopes the result**: the firm advance is the R-2 perturbation-artifact correction; the register-trace normalization $\kappa^2 = A/Q$ is DERIVED-CONDITIONAL on the register-scalar density $\rho = I_Q/Q$. Locked inputs: $\delta_X = 5/19$ (truncated octahedron, O_h), $\delta_Y = 7/23$ (truncated icosahedron, I_h), $A = \delta_X \delta_Y = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$.

§2. Heat-kernel setup and the finite-register heat-trace moments

The 1-loop effective action is $\Gamma = \frac{1}{2} \text{Tr}_C \log L = -\frac{1}{2} \int_0^\infty \frac{dt}{t} \text{Tr}_C e^{-tL}$, with

$\text{Tr}_C e^{-tL} = a_0 - a_1 t + a_2 t^2 - \dots$, $a_0 = \dim$, $a_1 = \text{Tr}_C L$, $a_2 = \frac{1}{2} \text{Tr}_C (L^2)$. Writing $L = L_0 + \kappa V$ (V the unit rank-1 β_0 -selected structure coupling β_0 to the nine target modes — X triplet, Y triplets T_{1u}, T_{2u} — with the Z_2 -odd mode decoupled):

$a_0 = \dim = Q = 11$ (mode count = leading rank coeff), $a_1 = \text{Tr}_C L = 13.30$ ($\Delta a_1 = 0$, coupling – immunity).

Correction (external review 1.2): the mode count is $a_0 = \dim = Q$, the leading coefficient — not a_1 . $a_1 = \text{Tr}_C L = 13.30 \neq Q$; its coupling-immunity ($\Delta a_1 = 0$, since V is off-diagonal) is a *separate* fact that preserves the mode count but does not fix the democratic distribution. The “/ Q ” is the **normalized trace** $\tau_Q = Q^{-1} \text{Tr}_C$ (with $Q = a_0$). (Verified: $a_0=Q=11$, $a_1=\text{Tr} \neq Q$, τ_Q normalization.)

§3. R-1: the Register-Trace Normalization Theorem (M44.T1)

Theorem M44.T1 (Register-Trace Normalization). [DERIVED-CONDITIONAL] *If the register-scalar coupling A distributes over the register via the maximally-mixed density $\rho = I_Q/Q$ (the register-trace normalization), then the per-mode cross-coupling is $\kappa^2 = A/Q$, with $Q = a_0 = \dim$ (the normalized-trace rank) and $A = \delta_X \delta_Y$ (Gilkey product).*

Proof (conditional). (i) The normalized trace on C^Q is $\tau_Q = Q^{-1} \text{Tr}_C$, with rank $a_0 = Q$. (ii) By the Gilkey product theorem on $\Gamma_X \otimes \Gamma_Y$ (using $[su(2)_X, su(2)_Y] = 0$), the effective coupling is $\xi_{eff} = \delta_X \delta_Y = A$ (PROVEN); it multiplies the register norm $|\Phi|^2$ in ZS-F1's $(1 + A|\Phi|^2)R$. (iii) Assuming A acts through the maximally-mixed density $\rho = I_Q/Q$, the single-mode cross-coupling is $\kappa^2 = A\langle a|\rho|a \rangle = A/Q$. ; ◻

The undischarged step is (iii): $\rho = I_Q/Q$ is the *symmetry-natural* / maximal-entropy state, but is **not**

derived from the parent action. So M44.T1 closes the *arithmetic* of the register-trace distribution, not a full action-level heat-kernel derivation.

The “/Q” is the normalized-trace rank $a_0 = Q$ — well-defined and exact. What remains conditional is the *democratic distribution* ($\rho = I_Q/Q$): why A is spread uniformly rather than sector-weighted. This is the genuine content, addressed (within irreps, and corrected of a false counter-signal) in §5. (*Verified: $\kappa^2 = A/a_0 = A/Q$ arithmetic, $\rho = I_Q/Q$ asserted not derived.*)

§4. Consistency: $\Delta a_2 = 9\kappa^2$

For the rank-1 β_0 -selected structure, $\text{Tr}(V^2) = 18$ (nine off-diagonal pairs), so

$\Delta a_2 = 9\kappa^2 = 9A/Q = 315/4807 = 0.06552943 \dots$, matching ZS-M6 (4.3.1) exactly; the factor $9 = 3 \times 3$ is (three target irreps) $\times$ (dimension 3), both PROVEN (ZS-F2 §4.2A). ($\Delta a_2 = 9\kappa^2$ is a *forward consequence* of κ^2 and $\text{Tr}(V^2) = 18$; the $a_2 = \frac{1}{2}\text{Tr}(L^2)$ convention is unaffected by the a_0/a_1 correction.) (*Verified: $\Delta a_2 = 9\kappa^2 = 315/4807$.*)

§5. R-2: the register-scalar embedding (within-irrep resolved, cross-block open)

§5.1 The split

The register-scalar assumption R-2 — that A acts as A, I_Q — decomposes into two logically distinct claims:

- **R-2(a) within-irrep:** the coupling is constant on each target irrep (no splitting within a triplet).
- **R-2(b) cross-sector:** the *same* κ^2 applies to all sectors ($\kappa_X^2 = \kappa_{Y_1}^2 = \kappa_{Y_2}^2$), i.e. $g_\Gamma^2 = \dim(\Gamma)\kappa^2$ is democratic — as opposed to sector-weighted ($\kappa_\Gamma^2 \propto \delta_{\text{sector}}$).

R-1 (§3) supplies the register *average* A/Q ; it does not by itself fix the per-sector breakdown, which is R-2(b).

§5.2 R-2(a) — within-irrep constancy [DERIVED]

The register carries the polyhedral symmetries $O_h(X)$ and $I_h(Y)$. The non-minimal coupling

$(1 + A|\Phi|^2)R$ is built from the G -invariant register norm $|\Phi|^2$ and the G -invariant register curvature, hence the coupling operator is G -equivariant. By **Schur’s lemma**, a G -equivariant operator restricted to an irreducible representation is a scalar. All three targets are **dimension-3 irreps** (T_1 of O_h ; T_{1u}, T_{2u} of I_h ; ZS-F2 §4.2A Adjoint Obstruction, PROVEN). Therefore the coupling is constant within each target — R-2(a) is **DERIVED**.

Crucial limitation (review error 4). Schur equivariance under $O_h \times I_h$ gives only *block* scalarity: the density is forced to the form $\rho = c_Z I_Z \oplus c_X I_X \oplus c_{Y_1} I_{Y_1} \oplus c_{Y_2} I_{Y_2}$ (one constant per irreducible block), **not** the equality $c_Z = c_X = c_{Y_1} = c_{Y_2}$ that $\rho = I_Q/Q$ requires. Closing the latter needs one of: (i) a larger symmetry acting transitively across blocks, (ii) a unique KMS state, (iii) an entropy-extremizing

state-selection action, or (iv) an explicit Regge-Hessian computation showing all block weights equal (§6.3). None is supplied here — so the *cross-block* equality is the residual OPEN item, not a Schur consequence. (*Verified: dim-3 irreps => Schur scalar; block weights c_Γ not forced equal.*)

§5.3 R-2(b) — democratic across sectors [CONFIRMED-NUMERICALLY, exact diagonalization]

R-2(b) is examined by inverting the ZS-M6 §2.3 spectrum exactly (a consistency check on the democratic ansatz). Because β_0 couples only to the three targets, the coupled modes live in a 4×4 **arrowhead** submatrix with β_0 -diagonal 1 and target diagonals $d = (2.0556, 2.2778, 2.7658)$; its four eigenvalues are the shifted modes $\mu = (0.9517, 2.0736, 2.2952, 2.7787)$. The arrowhead inverse-eigenvalue formula recovers the physical couplings **exactly** (no perturbation theory):

$$c_\Gamma^2 = g_\Gamma^2 = - \frac{\prod_k (\mu_k - d_\Gamma)}{\prod_{j \neq \Gamma} (d_j - d_\Gamma)}.$$

The result:

target	g_Γ^2/κ^2 (exact)	democratic	sector-weighted
X	2.996	3.000	$3\delta_X/\delta^-$
Y_1	2.989	3.000	3.000
Y_2	3.021	3.000	3.000

The recovered couplings are democratic to a **0.45% spread** — at the 4-decimal input precision of the §2.3 spectrum. The decisive discriminator: $c_X^2/c_{Y_1}^2 = 1.002$, matching democratic (1.000) and **excluding** the sector-weighted prediction $\delta_X/\delta_Y = 0.865$ by 13.5%. So R-2(b) is **democratic at the consistency level** (the sector-weighted 0.865 is excluded *given* the arrowhead structure). This does **not** by itself close R-2: an action-independent forward derivation, and the global operator $\hat{R} \propto I_Q$ (§6), remain required.

(*Verified: $g_\Gamma^2/\kappa^2 \approx 3$, spread 0.45%, $c_X^2/c_{Y_1}^2 = 1.002$ not 0.865; consistency-level.*)

§5.4 Correction to ZS-M6 §10.4 [honest amendment]

ZS-M6 §10.4 reported $g_X^2/\kappa^2, g_{Y_1}^2/\kappa^2, g_{Y_2}^2/\kappa^2 \approx (2.61, 3.05, 3.13)$ — a “~15% asymmetry” attributed to

“4-decimal rounding.” The exact solve identifies the precise mechanism: those numbers come from the

2nd-order perturbation formula $g^2 = \Delta\lambda(\lambda - \lambda_{\beta_0})$, which at this coupling strength ($|\Delta\lambda_{\beta_0}| = 0.048$)

carries a ~ 13 error on the X channel. Reproducing that formula gives (2.61, 3.05, 3.13) and

$c_X^2/c_{Y_1}^2 = 0.855$ — which **spuriously coincides with** $\delta_X/\delta_Y = 0.865$. This is a genuine trap: the

perturbation artifact *looks like* sector-weighting. The exact arrowhead solve removes the artifact and yields the democratic (2.996, 2.989, 3.021). The amendment strengthens ZS-M6’s conclusion

(democratic $g^2 = 3\kappa^2$) while correcting its diagnosis (perturbation error, not merely rounding, and a δ_X/δ_Y -mimicking trap). (*Verified: 2nd-order reproduces 2.61, 2nd-order ratio 0.855 mimics dX/dY .*)

§5.5 The 50-digit closure [CONFIRMED-NUMERICALLY, residual eliminated]

The v1.1 confirmation was limited to 0.45% by the 4-decimal §2.3 input. This is now removed. Build L at 50-digit precision from ingredients derived **independently of the spectrum**: the sector eigenvalues from the polyhedra graph Laplacians ($\lambda_X = 19/18, \lambda_{Y_1} = 23/18, \lambda_{Y_2} = \frac{5-\sqrt{5}}{2} \cdot \frac{23}{18}$); the magnitude $\kappa^2 = A/Q$ (§3, normalized trace + Gilkey); the democratic per-target norm $g_\Gamma^2 = \dim(\Gamma)\kappa^2 = 3\kappa^2 = 105/4807$ (§5.2, Schur + Dimensional Coupling Norm); and the rank-1 β_0 -selected structure (§6, R-3).

Diagonalizing the resulting arrowhead at 50 digits gives, at $\mu = 1$:

$$\lambda = 0.95169951 \dots, 2.07357649 \dots, 2.29524490 \dots, 2.77865788 \dots,$$

which rounds **exactly** to the ZS-M6 §2.3 values 0.9517, 2.0736, 2.2952, 2.7787 on all four modes.

Inverting this 50-digit spectrum via (5.1) recovers

$$g_X^2/\kappa^2 = g_{Y_1}^2/\kappa^2 = g_{Y_2}^2/\kappa^2 = 3.000 \dots \quad (\max \text{ deviation } 2.2 \times 10^{-58}).$$

So the exactly-democratic coupling **reproduces §2.3** to 50 digits, and the v1.1 0.45% spread was the 4-decimal rounding of §2.3. **Honest limitation (external review 1.3)**: this is a *consistency check*, not a fully action-independent forward derivation. The construction *inputs* the democratic per-target norm $g_\Gamma^2 = 3\kappa^2$; and the ZS-M6 §2.3 target spectrum may itself have been generated with democratic coupling.

A genuinely independent forward computation would build L from the ZS-F1/S14 action and the polyhedral incidence matrices, compute each coupling *without* the democratic ansatz, and only then forward-diagonalize. That computation is **not** performed here. What is established: the democratic ansatz is *internally consistent* with §2.3 to machine precision, and the sector-weighted alternative is excluded *given* the arrowhead structure. (*Verified: 50-digit spectrum rounds to 2.3, inversion $g^2/\kappa^2=3$, positive-definite; consistency-level, not independence.*)

Deepest item — see §6. The one item beyond §5 (an explicit lattice/Regge realization of the register-scalar operator $\hat{R} \propto I_Q$) is treated in the integrated §6 (Regge-cell realization). Result: the primitive-cell deficit is sector-uniform (§6.1, DBI), but the global operator equality remains **OPEN** (§6.3, the Regge-Hessian gate). So $\rho_Q = I_Q/Q$ and hence $\kappa^2 = A/Q$ remain DERIVED-CONDITIONAL.

§6. Regge-cell realization (integrated from ZS-M45)

This section integrates ZS-M45 v1.1. It addresses the same residual as §5: whether the register-scalar density $\rho_Q = I_Q/Q$ (equivalently $\hat{R} \propto I_Q$) can be grounded geometrically, rather than assumed. The honest result is a **partial** advance: local uniformity holds; the global operator equality does not follow.

§6.1 Uniform primitive-cell deficit [DERIVED-BY-INHERITANCE]

The ZS-M3 Regge-Holonomy framework gives the holonomy deficit per primitive register cell as $\delta\varphi_{cell} = A \cdot I_{cell}$ with

$$I_{cell} = \frac{d_v}{r} = \frac{4}{4} = 1, \quad \delta\varphi_{cell} = A = \frac{35}{437},$$

where $d_v = 4$ is the **valence** of the $Q = 11 \leftrightarrow 4$ -valent $j = \frac{1}{2}$ tetrahedron and $r = 4$ is the ZS-F4 structural constant (both sector-independent; the symbol d_v replaces the earlier clashing κ). Hence **every**

primitive register cell carries the identical local deficit A , independent of sector label X/Y . This is DERIVED-BY-INHERITANCE from ZS-M3 (inheriting its [DERIVED] status and primitive-cell scope).
(Verified: $d_{v/r}=1$, $\delta\phi_{cell}=A$, sector-independent.)

§6.2 Why local uniformity is insufficient [the gap]

The Regge curvature *operator* is **not** the deficit alone. In Regge calculus,

$$\hat{R} = \sum_c \delta_c w_c \Pi_c,$$

with hinge/dual-volume weights w_c and cell-incidence projectors Π_c . Uniform $\delta_c = A$ gives

$$\hat{R} = A \sum_c w_c \Pi_c, \text{ which is } \propto I_Q \text{ only if additionally}$$

$$w_c = \text{const} \quad \text{and} \quad \sum_c \Pi_c \propto I_Q.$$

Because the X and Y polyhedral cells have **different incidence structure**, “same local deficit” and “same global matrix element” $\langle \beta_0 | \hat{R} | X \rangle = \langle \beta_0 | \hat{R} | Y \rangle$ are **distinct claims**. M45’s §6.1 closes the first; the second does not follow.

§6.3 Graph-Laplacian coordinate diagnostic [COMPUTED-DIAGNOSTIC; genuine gate OPEN]

The decidable form of the residual is an explicit finite computation: construct the sector cell–hinge incidence matrices B_X, B_Z, B_Y and the diagonal hinge-area / dual-volume weight matrices W ; form the

register Regge Hessian $H_{\text{Regge}} = B^T W_{\text{hinge}} B$; and, with isotypic projectors $P_X, P_{Y_1}, P_{Y_2}, P_{\beta_0}$, test

$$h_X = h_{Y_1} = h_{Y_2} \text{ (diagonal) and } c_X = c_{Y_1} = c_{Y_2} \text{ (coupling).}$$

What is computed here (a diagnostic, not the gate). On the actual corpus mediators — truncated octahedron ($X, V = 24, \delta_X = 10/38 = 5/19$) and truncated icosahedron ($Y, V = 60,$

$\delta_Y = 28/92 = 7/23$) — the graph Laplacian $L = D - A$ and the coordinate vectors give **coordinate Rayleigh quotients**

$$h_X^{\text{coord}} = \frac{x^T L_X x}{x^T x} = 0.600, \quad h_Y^{\text{coord}} = 0.244277.$$

Crucial honest caveat (third review). The coordinate 3-spaces are **not** invariant eigenspaces of the graph Laplacian — the residuals are nonzero, $|L_X x - h_X^{\text{coord}} x|/|x| = 0.200$ and

$$|L_Y x - h_Y^{\text{coord}} x|/|x| = 0.050 \text{ — so (6.5) are **Rayleigh quotients, not eigenvalues**, and (6.5) is **not** a}$$

$B^T W_{\text{hinge}} B$ computation. (For reference, the true lowest triple eigenvalues are $\lambda_X = 2 - \sqrt{2} = 0.5858$ and $\lambda_Y = 0.2434$ — which also differ, so the diagnostic points the same way, but this remains a

graph-Laplacian statement, not the register Regge Hessian.) Building the genuine gate requires the hinge/cell incidence, an adopted discrete Hodge-star / dual-volume convention, true isotypic projectors, repeated-irrep multiplicity resolution, off-block mixing, and the mediator coupling block — a separate computation, **not performed here**.

What the diagnostic does and does not show. It is *diagnostic evidence* that the sector diagonals differ; it is **not** a proof that the physical register operator $\hat{R} \propto I_Q$ fails. The coupling democracy depends on the (undetermined) register measure. Two **benchmark** measures — not an exhaustive dichotomy — bracket the possibilities:

- **Mode-count (uniform-cell) benchmark** — the ZS-M3 Lemma 8.1 primitive cell is the uniform $j = \frac{1}{2}$ tetrahedron and both mediators are **valence-3** (equal per-vertex incidence): coupling **democratic**.
- **Metric (dual-volume) benchmark** — an (implementation-dependent) dual-volume weighting gives $w_x \approx 0.471$, $w_y \approx 0.921$, ratio ≈ 1.955 : coupling **sector-weighted** $\sim 2 \times$. (*No discrete Hodge-star / dual-cell normalization is derived for these numbers, so they are an implementation-dependent diagnostic, not a COMPUTED weight.*)

Other measures (Gibbs $\rho_\beta = e^{-\beta H}/Z$, sector-weighted traces, heat-kernel-regularized, modular/KMS weights) are equally admissible. So the honest statement is: **the register-scalar normalization has been narrowed to the action-level selection of a register measure**, with mode-count and metric as two explicit benchmarks. Under the mode-count measure — for which the a_0 leading heat-trace coefficient (each mode counted once, independent of eigenvalue) is a genuine physical argument — $\rho_Q = I_Q/Q$ and $\kappa^2 = A/Q$ hold; but the measure is **not derived from the parent action**, so the gate is **OPEN**. The anti-numerology check confirms the 1.955 factor is a genuine geometric ratio, **not** a corpus constant ($\neq \delta_y/\delta_x = 1.156$, $\neq$ any clean A, Q combination — flagged, not claimed). (*Verified: zs_m44_regge_verify.py 12/12 — valence-3 both, coordinate Rayleigh $\neq$ eigenvalue (residuals $\neq 0$), true triples 2 — $\sqrt{2}/0.2434$, dual-volume ratio 1.955 implementation-dependent, measures non-exhaustive.*)

§7. R-3: rank-1 β_0 -selection [DERIVED-CONDITIONAL]

ZS-M6 §7A (Sector Separation Theorem) proves $L_{XY} \equiv 0$ to all orders in the continuum QFT (non-minimal coupling generates X–Z and Z–Y but no direct X–Y; anomaly-free Ward identity). With $L_{XY} = 0$ and β_0 the unique Z-Spin mediator, Schur fixes the rank-1 β_0 -selected structure — but only **provided** the intertwiner spaces $\text{Hom}_G(V_X, V_Z)$ and $\text{Hom}_G(V_Z, V_Y)$ are each one-dimensional. That multiplicity has not been computed here; the Z_2 -odd mode being decoupled is supporting but not conclusive evidence. Status: **DERIVED-CONDITIONAL on Hom-multiplicity-one**. (*Verified: $L_{XY}=0$ + Schur; Hom-mult-1 required, uncomputed.*)

§8. Net status and hand-off to ZS-F36

Net epistemic status (the honest summary of this merged paper).

Claim	Status
$a_0 = Q$ (mode count), $a_1 = \text{Tr}, L \neq Q$	PROVEN
perturbative sector-asymmetry artifact removed ($0.855 \approx \delta_x/\delta_y$ is spurious)	PROVEN

Claim	Status
within-irrep scalarity (R-2a)	DERIVED
primitive-cell deficit uniformity $\delta\varphi_c = A$ (§6.1)	DERIVED-BY-INHERITANCE
coordinate Rayleigh diagnostic: sector coordinate-quotients differ (0.600 vs 0.244, §6.3)	COMPUTED-DIAGNOSTIC (not eigenvalues)
genuine register Regge Hessian $B^\top W_{\text{hinge}} B$	OPEN (not built: needs Hodge-star, isotypic projectors)
$\rho_Q = I_Q/Q / \hat{R} \propto I_Q$	OPEN , narrowed to action-level <i>register-measure selection</i> ; a_0 argues for mode-count
R-3 rank-1 β_0 -selection	DERIVED-CONDITIONAL (Hom-mult-1)
$\kappa^2 = A/Q$	DERIVED-CONDITIONAL (register-trace normalization)
ZS-F36 $\tilde{G}_s = 1$	DERIVED-CONDITIONAL (register-trace normalization)

Honest net status (prose). R-2(a) (within-irrep constancy) is DERIVED (Schur). The R-2 *perturbation-artifact correction* (§5.4) is the firm advance. R-1 and R-2(b) **merge** into the single conditional step — that A distributes via $\rho = I_Q/Q$ (register-trace normalization) — which is

Schur-constrained within irreps but not derived across sectors from the action, and whose lattice realization (M45) is incomplete (§5.5). Therefore

$$\kappa^2 = \frac{A}{Q} = \frac{35}{4807}, \quad g_{\text{reg}}^2 = \dim(Y)\kappa^2 = \frac{6A}{Q} = \frac{210}{4807} \quad [\text{DERIVED} - \text{CONDITIONAL on register} - \text{trace normal}.$$

with the value fixed by the LOCKED ZS-F1 action, $Q = 11$, and the polyhedra. ZS-F36 §6 uses

$g_{\text{reg}}^2 = 6A/Q$; accordingly $\tilde{G}_s = 1$ is **DERIVED-CONDITIONAL on register-trace normalization** — the register-scalar density $\rho = I_Q/Q$, whose action-level derivation remains OPEN; §6.3 provides only a graph-Laplacian diagnostic.

Honest scope. This does not touch the ZS-F36 **metric-scale no-go**: M_{UV} remains PROVEN-irreducible.

R-1/R-2/R-3 concern only the *dimensionless* coupling κ^2 ; the scale M_{UV} is orthogonal and unaffected. “

$\tilde{G}_s = 1$ DERIVED” therefore means the *normalization* is fixed, not that the physical odd-form stiffness is a pure number — it carries M_{UV} dimensions set elsewhere.

§9. Claim ledger

#	Claim	Status	Conditions
§2	$a_0 = \dim = Q$ (mode count); $a_1 = \text{Tr}, L \neq Q$, coupling-immune	PROVEN	corrects v1.1
T1	$\kappa^2 = A\langle a \rho_Q a\rangle = A/Q$	DERIVED-CONDITIONAL	$\rho_Q = I_Q/Q$
§4	$\Delta a_2 = 9\kappa^2 = 315/4807$	PROVEN	forward from T1
§5.2	R-2(a) within-irrep constancy	DERIVED	Schur + O_h/I_h
§5.2	cross-block equality $c_z = c_x = c_{Y_1} = c_{Y_2}$	OPEN	not a Schur consequence (review 4)
§5.3/5.5	R-2(b) cross-sector democratic ($g_\Gamma^2 = 3\kappa^2$)	CONFIRMED-CONSISTENCY	not action-independent (review 1.3)

#	Claim	Status	Conditions
§5.4	ZS-M6 §10.4 asymmetry = 2nd-order artifact	PROVEN	exact vs perturbative
§6	Regge-cell deficit uniformity $\delta\varphi_c = A$	DERIVED-BY-INHERITANCE	ZS-M3 (integrated M45)
§6.3	global operator $\hat{R} \propto I_Q$ (Regge Hessian)	OPEN	X/Y incidences differ
§7	R-3 rank-1 β_0 -selection	DERIVED-CONDITIONAL	§7A + Schur + Hom-mult-1 uncomputed
§7	$\kappa^2 = A/Q, g_{reg}^2 = 6A/Q$	DERIVED-CONDITIONAL	register-trace normalization
§7	ZS-F36 $\tilde{G}_s = 1$	DERIVED-CONDITIONAL	register-trace normalization
§5.5	50-digit recompute consistent with §2.3	CONFIRMED-CONSISTENCY	inputs democratic ansatz

§10. Falsification gates

- **F-M44.1.** $\Delta a_1 \neq 0$ (mode count not immune) $\Rightarrow$ R-1 fails. *PASS*.
- **F-M44.2.** Exact arrowhead solve giving $c_x^2/c_{Y_1}^2$ near $\delta_x/\delta_Y = 0.865$ rather than 1 $\Rightarrow$ coupling sector-weighted, R-2(b) fails. *PASS (exact ratio = 1.002)*.
- **F-M44.3.** If the 50-digit recompute (§5.5) gave g_r^2/κ^2 departing from 3 $\Rightarrow$ R-2(b) reopens. *PASS (recovered 3 to 2.2×10^{-58})*.
- **F-M44.4.** ZS-M6 §7A not establishing $L_{xy} = 0$ to all orders $\Rightarrow$ R-3 reopens. *PASS (inherited)*.
- **F-M44.5.** A claim that the §10.4 numbers (2.61, ...) prove sector-weighting is void: they are a perturbation artifact (§5.4).

§11. Anti-numerology

Every number is either PROVEN, exactly computed, or explicitly flagged conditional:

$A = \delta_x \delta_Y = 35/437$ (PROVEN), $Q = 11$ ($a_0 = \dim$, PROVEN), $\Delta a_2 = 9A/Q$ (exact), $Tr(V^2) = 18$ (exact); $\kappa^2 = A/Q$ and $g_r^2 = 3\kappa^2$ are **DERIVED-CONDITIONAL** on the register-trace normalization $\rho_Q = I_Q/Q$. **Nothing is fitted.** Crucially, this paper *removes* a latent numerical trap: the §10.4 ratio $0.855 \approx \delta_x/\delta_Y$ is a perturbation artifact, not a geometric law — reading it as sector-weighting would have been the numerology. The exact solve discriminates democratic (1.002) from sector-weighted (0.865) cleanly. The 0.45% residual is stated as input-precision, not concealed.

§12. Cross-version safety

- $(A, Q, \dim Z) = (35/437, 11, 2)$, $\kappa^2 = 35/4807$, $g_{reg}^2 = 210/4807$: **unchanged** (only epistemic status improves).

- **ZS-M6:** no value modified. **§10.4 amended** (the firm result): the “~15% asymmetry” is a 2nd-order perturbation artifact, not sector-weighting; the democratic $g^2 = 3\kappa^2$ conclusion is *strengthened* at the consistency level. R-1/R-2 reduce to the single register-trace normalization (DERIVED-CONDITIONAL); R-3 DERIVED-CONDITIONAL on Hom-mult-1. **No caveat is claimed unconditionally closed.**
- **ZS-F1/F2/S1:** inherited inputs (register-scalar action; Gilkey product; Mode-Count Collapse); no status changed.
- **ZS-F36:** $\tilde{G}_s = 1$ is **DERIVED-CONDITIONAL on register-trace normalization** (F36 v2.1 §6); metric-scale no-go untouched.
- **ZS-M45:** its content is now §6 of this paper; the standalone M45 v1.0–v1.1 are archived as internal companion drafts.

§13. Verification

zs_m44_verify_v1_2.py (19/19) and zs_review_verify.py (14/14) PASS. The review suite verifies the corrections: $a_0 = \dim = Q$ while $a_1 = Tr, L = 13.30 \neq Q$; τ_Q normalization; $\rho = I_Q/Q$ asserted (not action-derived); $\Delta a_2 = 9\kappa^2$ unaffected; the arrowhead/50-digit results are consistency-level; and the M45 operator-equality is OPEN. The §10.4 perturbation-artifact correction (democratic vs the 0.855 trap) is the firm, independently-checkable result.

§14. Acknowledgements & Code Availability

zs_m44_verify_v1_2.py (19/19), zs_review_verify.py (14/14), and zs_m44_regge_verify.py (12/12, the §6.3 coordinate diagnostic) reproduce all checks. AI tools (Anthropic Claude) were used for verification and drafting; the author assumes full responsibility, including the honest §10.4 amendment and the explicit §5.5 residual (rather than an over-claimed unconditional closure).

§15. Appendix A — Deep-exploration (issue-tree) record

Step 0 (long list, 7). (1) does R-2 split into within-irrep and cross-sector parts? (2) does Schur + polyhedral symmetry close within-irrep? (3) is cross-sector democratic or sector-weighted? (4) can exact diagonalization decide it, avoiding perturbation error? (5) what is the ZS-M6 §10.4 asymmetry, really? (6) does R-2(b) follow from R-1’s mode count? (7) any residual after closure?

Step 1 (MECE). I1 R-2(a) within-irrep $\{(1),(2)\}$. I2 R-2(b) cross-sector $\{(3),(4),(6)\}$. I3 the §10.4 diagnosis + residual $\{(5),(7)\}$.

Step 2–3 (tree + status). I1 $\rightarrow$ I2 $\rightarrow$ I3. I1 **DERIVED** (Schur on dim-3 irreps). I2 **CONFIRMED democratic** by exact arrowhead solve (0.45%, excludes 0.865); note R-1 gives only the *average*, so the exact per-sector solve is load-bearing, not decorative. I3: the §10.4 asymmetry is a **2nd-order perturbation artifact** mimicking δ_x/δ_y — corrected; the verification residual is then **eliminated** by the v1.2 50-digit recompute (democratic reproduces §2.3, inverts to 3 at $< 10^{-45}$), leaving only the narrow lattice- $\hat{R}$ completion.

Step 4 (convergence). Re-traversal change counts [5, 1, 0] decreasing $\Rightarrow$ converged. The node that flipped on re-traversal was I2: the *first* pass (2nd-order perturbation, as in the prior exploration) suggested sector-weighting (ratio 0.855); the exact arrowhead solve on re-traversal corrected this to democratic (1.002). This flip is the paper’s central lesson — the discriminator must be exact, not perturbative.

Step 5 (value). The firm advance is the §10.4 perturbation-artifact correction (R-2(a) DERIVED; the $0.855 \approx \delta_x/\delta_y$ trap defused). R-1 and R-2(b) merge into the single **register-trace normalization** step ($\rho = I_Q/Q$), which is DERIVED-CONDITIONAL — Schur-constrained within irreps, but not action-derived across sectors, and with an incomplete lattice realization (M45, review 1.4). Honest terminus: $\kappa^2 = A/Q$ **DERIVED-CONDITIONAL**, not fully discharged — the a_0/a_1 correction and the consistency-vs-independence distinction (review 1.2–1.4) are integrated.

§16. References

(ZS-M45 v1.1 is integrated into §6 of this paper; the standalone drafts are archived as internal companion notes.)

1. P. B. Gilkey, *Invariance Theory, the Heat Equation, and the Atiyah–Singer Index Theorem*, 2nd ed., CRC Press (1995).
2. D. V. Vassilevich, *Heat kernel expansion: user’s manual*, Phys. Rept. **388** (2003) 279, arXiv:hep-th/0306138.
3. J.-P. Serre, *Linear Representations of Finite Groups*, GTM **42**, Springer (1977). [Schur]
4. G. H. Golub, *Some modified matrix eigenvalue problems*, SIAM Review **15** (1973) 318. [arrowhead inverse-eigenvalue]
5. ZS-F1, *The Non-Minimal Coupling* $(1 + A|\Phi|^2)R$ (Z-Spin corpus).
6. ZS-F2, *Product Structure* $A = \delta_x \delta_y$; *Adjoint Obstruction* (Z-Spin corpus).
7. ZS-M2, $[su(2)_x, su(2)_y] = 0$ *Sector Independence* (Z-Spin corpus).
8. ZS-M6, *Block-Laplacian; Register-Total Normalization; Dimensional Coupling Norm; §7A Sector Separation* (Z-Spin corpus).
9. ZS-F36 v2.1, *Integral UV Normalization of the Z-Spin Odd Three-Form* (Z-Spin corpus).

§17. Version History

- **Version:** v1.6 (July 2026, finalized for public release — supersedes v1.5–v1.0; no research change). Editorial finalization only: confirms the three proofread items (ledger **T1** status/condition filled — $\kappa^2 = A\langle a|\rho_Q|a\rangle = A/Q$ DERIVED-CONDITIONAL, $\rho_Q = I_Q/Q$; the §8 wording “*whose action-level derivation remains OPEN; §6.3 provides only a graph-Laplacian diagnostic*”; §6.3 = COMPUTED-DIAGNOSTIC), unifies the mediator terminology to **Z-Spin mediator** (β_0 ; the actor of mediation is the Spin), and updates the ZS-F36 cross-reference **v2.0** $\rightarrow$ **v2.1**. 19/19 + 14/14 + 12/12 PASS; no numerical value changed; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.
- **v1.5** (supersedes v1.4–v1.0; integrates **ZS-M45 v1.1** as §6; §6.3 correctly reclassified as a graph-Laplacian *coordinate diagnostic*, not a Regge-Hessian theorem) — the two papers address

one problem: M45 is the geometric lemma for M44's final normalization step, so they are merged). Second external review integrated (body synchronized to the conditional conclusions). **v1.5 corrects a v1.4 over-claim (third review):** the $h_x = 0.600$, $h_y = 0.244$ values are graph-Laplacian **coordinate Rayleigh quotients**, *not* Regge-Hessian eigenvalues (the coordinate 3-space is not Laplacian-invariant: residuals 0.200, 0.050 $\neq 0$; the true lowest triples are $2 - \sqrt{2} = 0.5858$ and 0.2434). So §6.3 is a **COMPUTED-DIAGNOSTIC**, not a theorem: it gives *diagnostic evidence* that the sector diagonals differ, but the genuine register Regge-Hessian $B^\top W_{\text{hinge}} B$ (hinge incidence + discrete Hodge-star + isotypic projectors) is **not** built here and remains **OPEN**. The residual is *narrowed* to the action-level selection of a register measure (mode-count and metric being two benchmark choices, not an exhaustive dichotomy). **This version corrects two over-claims of the earlier drafts:** (i) the finite-matrix heat trace has $a_0 = \dim = Q$ as the *leading rank coefficient / mode count* — **not** a_1 (which is $\text{Tr}, L \neq Q$); the “/Q” is the **normalized trace** $\tau_Q = Q^{-1} \text{Tr}$, and the maximally-mixed density $\rho = I_Q/Q$ is *asserted*, not derived from the action; (ii) the arrowhead inversion and 50-digit recompute are **consistency checks**, not a fully action-independent forward derivation. Consequently $\kappa^2 = A/Q$ is **DERIVED-CONDITIONAL on register-trace normalization**, and the genuine result is: *R-2 perturbative ambiguity corrected; normalized-trace bridge remains conditional*. v1.0 closed **R-1**; v1.1 closed **R-2** (Schur within-irrep + exact arrowhead democratic, 0.45% input-limited). **v1.2 eliminates the §5.5 residual: a 50-digit physical-action recompute** builds L from the independently-derived ingredients (polyhedra eigenvalues, $\kappa^2 = A/Q$ from §3, democratic $g_\Gamma^2 = 3\kappa^2$ from §5.2 — none from the spectrum), reproduces the ZS-M6 §2.3 spectrum exactly, and inverts to democratic to $< 10^{-45}$, proving the v1.1 “0.45% spread” was **purely §2.3's 4-decimal rounding**.

- **v1.5 (July 2026, current — §6.3 reclassified):** Third review correctly caught that the v1.4 values $h_x = 0.600$, $h_y = 0.244$ are graph-Laplacian **coordinate Rayleigh quotients**, not Regge-Hessian eigenvalues (coordinate 3-spaces are not Laplacian-invariant; residuals 0.200, 0.050 $\neq 0$; true triples $2 - \sqrt{2}$, 0.2434). §6.3 is downgraded from a “Regge-Hessian computation” to a **COMPUTED-DIAGNOSTIC**; the genuine $B^\top W_{\text{hinge}} B$ gate (hinge incidence, discrete Hodge-star, isotypic projectors) remains **OPEN**. “Mode-count vs metric” is reframed as **two benchmark measures, not an exhaustive dichotomy** (Gibbs/KMS/heat-kernel also admissible); the dual-volume numbers are flagged implementation-dependent (no Hodge-star convention derived). Also: R-3 stated DERIVED-CONDITIONAL in the abstract (consistent with §7); intro “v1.2 (this version)” fixed; ledger T1 cells filled; verification suite mention (12/12) added; reference 9 $\rightarrow$ F36 v2.0. 12/12 diagnostic checks; no numerical value changes; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.
- **v1.4 (July 2026 — executed §6.3, later reclassified in v1.5):** Performs the Regge-Hessian computation on the actual corpus polyhedra (truncated octahedron X , $V = 24$; truncated

icosahedron Y , $V = 60$). Reported the coordinate quotients $h_x = 0.600$, $h_y = 0.244$ as “provably unequal diagonal block weights” — **corrected in v1.5** (these are Rayleigh quotients, not eigenvalues) and $\rho_Q = I_Q/Q$ can only be the **mode-count measure**. The coupling democracy is thereby **localized** to one decidable choice — mode-count (uniform-cell, both valence-3 $\Rightarrow$ democratic) vs metric (dual-volume ratio 1.955 $\Rightarrow$ sector-weighted); the a_0 leading heat-trace coefficient argues for mode-count but does not derive it. Gate **not closed**, but sharply localized. 10/10 new checks; anti-numerology confirms 1.955 is a geometric ratio, not a corpus constant. No numerical value changes; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

- v1.3 (July 2026 — integrates ZS-M45 v1.1):** Merges the ZS-M45 Regge-cell lemma as §6 and fully synchronizes the body with the conditional conclusions (removes all residual v1.0/v1.1 closure language). Title dropped “Closing.” Corrects the $\kappa^2 = A/\tau_Q$ notation slip to $\kappa^2 = A\langle a|\rho_Q|a\rangle = A/Q$ (the normalized trace τ_Q is a functional, not a divisor). Adds the crucial limitation that $O_h \times I_h$ Schur equivariance forces only *block* scalarity $(c_Z, c_X, c_{Y_1}, c_{Y_2})$, **not** their equality — so $\rho_Q = I_Q/Q$ is OPEN at the action level. Re-scopes R-3 to DERIVED-CONDITIONAL on Hom-multiplicity-one. Reframes the finite-matrix Taylor coefficients as *finite-register heat-trace moments* (the discrete counterpart of continuum heat-kernel coefficients), not literal Seeley–DeWitt invariants. Adds the net-epistemic-status table (§8) and the explicit Regge-Hessian test (§6.3). 19/19 + 14/14 checks; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.
- v1.2 (July 2026):** Corrected the a_0/a_1 conflation (mode count is $a_0 = \dim = Q$, not $a_1 = \text{Tr}, L = 13.30$); re-scoped M44.T1 to DERIVED-CONDITIONAL ($\rho_Q = I_Q/Q$ asserted, not action-derived); flagged the arrowhead inversion and 50-digit recompute as consistency checks; the firm result is the §5.4 perturbation-artifact correction. (*Internal pre-review drafts also numbered v1.2/v1.3 — which over-claimed the ZS-M45 lattice- $\hat{R}$ as closing the register-scalar operator — are superseded and archived as internal drafts.*)
- v1.1 (July 2026):** Split R-2 into within-irrep (Schur) and cross-sector (arrowhead) parts; corrected the ZS-M6 §10.4 perturbation artifact. (*Erroneously called the “/Q” the “ a_1 mode count” — corrected in v1.2.*)
- v1.0 (July 2026):** Introduced the Register-Trace Normalization Theorem; reduced the residual to R-2; derived R-3 from §7A. 13/13 checks.